Lab 4 Practice: Direction Game 1

Description

In this lab, you will implement a direction game using a 4-bit Linear Feedback Shift Register (LFSR) and a single finite-state machine (FSM).

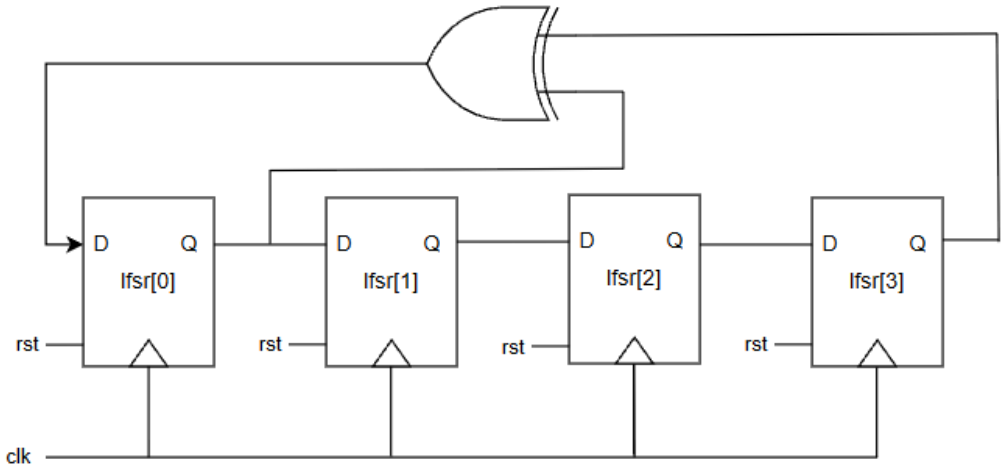
The LFSR generates a pseudo-random target direction for each round. The player must press the direction button that matches the current target. A correct input advances the game to the next round with a new target direction, while a wrong input ends the game immediately.

I/O List

I/O	Connected to	Definition
clk	W5	100 MHz clock signal.
rst	BTNC	Asynchronous active-high reset signal for both the LFSR and FSM.
btnL	BTNL	Left direction input.
btnR	BTNR	Right direction input.
btnD	BTND	Down direction input.
btnU	BTNU	Up direction input.
LED[3:0]	LD3 ~ LD0	Displays the LFSR state in the INIT and PLAY FSM states.
LED[4]	LD4	Always 1'b0 in the INIT and PLAY FSM states.
LED[6:5]	LD6 ~ LD5	Displays the target direction code in the PLAY FSM state.
LED[15:7]	LD15 ~ LD7	Always 9'b0 in the INIT and PLAY FSM states.

LFSR

Implement a 4-bit LFSR as shown in the following block diagram.



*Reset State: lfsr = 4'b0001

In the following description, lfsr[3:0] denotes the current state of the 4-bit LFSR.

- The LFSR must never enter the 4'b0000 state.
- LED[3:0] displays the current LFSR state (lfsr[3:0]) in the INIT and PLAY FSM states.
- The target direction of each round is determined by lfsr[1:0].
- The LFSR advances by exactly one state on each update.
  Example: 4'b0001 → 4'b0011 → 4'b0111

Direction Code

Direction	lfsr[1:0] (Target Code)	Button
Left	2'b00	btnL
Right	2'b01	btnR
Down	2'b10	btnD
Up	2'b11	btnU

Each button press should be treated as a single input event.

FSM

At any FSM state, rst == 1 asynchronously resets the FSM to the INIT state and the LFSR to 4'b0001.

INIT State

- LED[3:0] = lfsr, LED[4] = 1'b0, LED[6:5] = 2'b00, and LED[15:7] = 9'b0.
- Each btnR press advances the LFSR by exactly one state, allowing the player to select the starting LFSR state.
- When btnU is pressed, the FSM enters the PLAY state, and the current lfsr[1:0] becomes the target direction of the first round.

PLAY State

- The game continues to the next round after each correct input and ends immediately after a wrong input.
- LED[3:0] = lfsr, LED[4] = 1'b0, LED[6:5] = lfsr[1:0], and LED[15:7] = 9'b0.
- The target direction remains unchanged until the player presses a direction button.
- No input: Stay in the PLAY state. Nothing changes.
- Correct input: Advance the LFSR by exactly one state. The new lfsr[1:0] becomes the target direction of the next round.
- Wrong input: Go to the FINAL state immediately. The LFSR does not advance.

FINAL State

- Entering the FINAL state starts a 3-second timer.
- Use the following parameter to generate a one-second tick from the 100 MHz clock:

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parameter ONE_SEC = 27'd100_000_000;
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- LED[15:0] remains 16'hFFFF for the entire 3 seconds.
- All direction-button inputs are ignored, and the LFSR does not advance.
- After 3 seconds, the FSM automatically returns to the INIT state, and the LFSR is reset to 4'b0001.